ModelIt!

Student Activity Pack

Building a Space Elevator: The Ultimate Engineering Challenge

Modeling the Physics and Materials Science of a Tether to the Stars

Grade 10L3 | NGSS HS-PS2-2, HS-PS1-3 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think a space elevator would look like and how would it work?**

2. **Why do you think rockets are so expensive for reaching orbit, and could a cable be cheaper?**

3. **Draw the forces acting on a cable stretching from Earth's surface to space — what keeps it from falling down?**

4. **What material properties would a space elevator cable need and do any current materials qualify?**

2 Scoring Rubric

ModelIt!

> Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Cable Tensile Strength | Gravitational Force Gradient | Centrifugal Counterweight | Climber Mass | Orbital Altitude | Material Limits | Energy Consumption | Payload Capacity | Atmospheric Stress | Economic Viability

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Carbon Nanotube Ideal

Theoretical maximum tensile strength of perfect carbon nanotubes

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Current Materials

Best currently manufacturable cable material

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Climber Stress Test

Maximum economic payload capacity

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Orbital Debris Risk — The probability of the cable being struck by space debris or micrometeorites at various altitudes — the cable crosses all orbital zones and must either avoid or survive millions of potential impacts over its operational lifetime

Climber Power System — The method of transmitting energy to the ascending climber — ground-based lasers, microwave beaming, or onboard generators — each with different efficiency, weight, and safety trade-offs

Anchor Station Design — The engineering of the ground-level platform, likely a mobile ocean platform near the equator, that must maintain cable tension, handle climber launch and recovery, and survive weather extremes

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

Could we actually build a cable from the ground to space — and would it fundamentally change humanity's relationship with the cosmos? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that the space elevator is a materials science problem rather than a physics or engineering problem?**

2. **What does the strength-to-weight ratio gap reveal about the relationship between theoretical material properties and manufacturing capability?**

3. **If the materials challenge were solved tomorrow, what other variables in your model would become the new limiting factors?**

4. **How would a functioning space elevator change the economics and accessibility of space — and what societal changes might follow?**

5. **What are the risks of a space elevator failure — what happens if a 100,000 km cable breaks?**

8 STEM Challenge: Design a Space Elevator Materials Research Roadmap

ModelIt!

YOUR ENGINEERING MISSION

Design a 25-year materials science research roadmap that identifies the specific breakthroughs needed to close the gap between current carbon nanotube manufacturing capability and space elevator requirements.

THE SCENARIO

A consortium of space agencies and technology companies has committed $10 billion over 25 years to develop the materials needed for a space elevator. Your team must design the research roadmap: identifying the specific manufacturing challenges, the intermediate milestones, the testing protocols, and the decision points where the program should continue, pivot, or redirect to alternative materials.

Engineering Design Thinking

1. WHAT IS THE CURRENT STATE OF CARBON NANOTUBE FIBER MANUFACTURING AND WHAT SPECIFIC DEFECTS LIMIT STRENGTH?
2. WHAT MANUFACTURING PROCESSES SHOW THE MOST PROMISE FOR SCALING NANOTUBE QUALITY AND LENGTH SIMULTANEOUSLY?
3. WHAT INTERMEDIATE APPLICATIONS (BRIDGES, AIRCRAFT, BODY ARMOR) COULD DEMONSTRATE PROGRESS AND PROVIDE REVENUE DURING THE 25-YEAR DEVELOPMENT?
4. WHAT ALTERNATIVE MATERIALS (BORON NITRIDE NANOTUBES, GRAPHENE RIBBONS, DIAMOND NANOTHREADS) SHOULD RECEIVE PARALLEL RESEARCH INVESTMENT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Materials Scientists develop new materials with extreme properties for aerospace, energy, and infrastructure applications. They work at national laboratories, aerospace companies, universities, and materials startups, earning $80,000-$160,000/year. Space Systems Engineers who design orbital infrastructure and launch systems earn $90,000-$170,000/year.